

Percents Greater Than 100% and Less Than 1%

Lesson 4-3

Name: _____

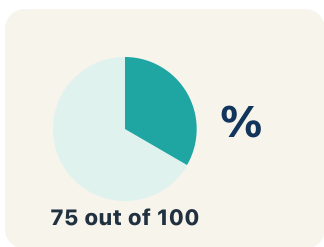
Date: _____

Class: _____

Key Vocabulary

Level 1 support

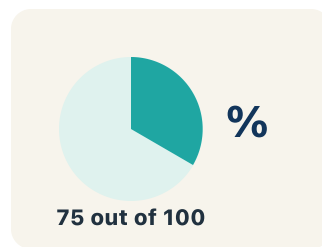
Picture first, then the word, then a plain-language meaning. Say each word out loud.



45% means 45 out of 100

Percent

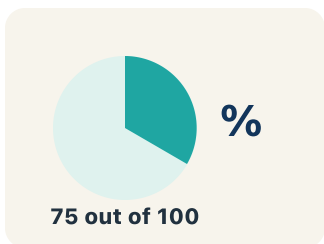
A way to compare a number to 100, shown with the % sign.



150% means 1.5 times the whole

Greater than 100%

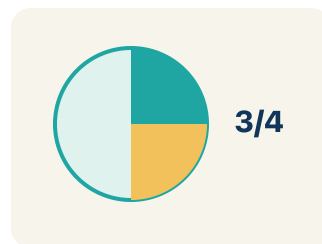
A percent bigger than 100%, more than the whole thing.



0.5% of 1,000 = only 5 items — a tiny sliver

Less than 1%

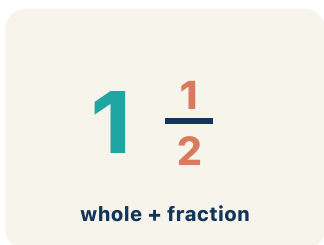
A percent smaller than 1%, a very tiny part.



150% = 1.5 = $\frac{3}{2}$

Equivalent

Having the same value, just written a different way.



150% = 1 and $\frac{1}{2}$ = 1.5

Mixed number

A whole number plus a fraction, like $1\frac{1}{2}$.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can write and interpret percents greater than 100% and less than 1%.

1 Notice

The arcade just posted its monthly leaderboard!

2 Key idea

I can write and interpret percents greater than 100% and less than 1%.

3 Words I'll use

Percent

Greater than 100%

Less than 1%

Equivalent

Mixed number



Think About It

- What does it mean when a score is shown as 150% of the target?
- Can a percent be less than 1%? What would that look like?
- How do you write a percent greater than 100% as a decimal or fraction?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

I do — watch

Follow each step as your teacher solves it.

Problem: Which of the following is greater than 100%?

- A. 0.75
- B. $\frac{3}{4}$
- C. 1.25
- D. $\frac{99}{100}$

Step 1 1.25 = 125%, which is greater than 100%.

Step 2 The other values are all less than 1 (less than 100%).



Answer: C. 1.25

 We do — together

Solve this one with your class using the same steps.

Problem: What is 200% written as a decimal?

- A. 2.0
- B. 0.2
- C. 20.0
- D. 0.02

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: What is 0.5% written as a decimal?

- A. 0.005
- B. 0.05
- C. 0.5
- D. 5.0

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The leaderboard shows one player hit 150% of the goal and a bonus round gives a 0.5% boost. How can a percent be more than the whole, and how can one be a tiny sliver?

Level 1 support

Start here: 150% is more than one whole, and 0.5% is a tiny piece smaller than 1%.

Need a hint?

Sentence starters (optional)

____% means ____ because it is more / less than 100%.

____% significa ____ porque es más / menos que 100%.

A percent over 100% means ____, and a percent under 1% means ____.

Un porcentaje mayor que 100% significa ____, y uno menor que 1% significa ____.

Word bank: percent greater than 100% less than 1% equivalent

Listen for: Students explain 150% is more than one whole (the player beat the goal) and 0.5% is far less than one whole (a tiny boost), connecting to the idea of 100% as the whole.

Level 2

100% means the whole goal. Explain why scoring 150% does NOT mean the player made an error, but scoring '150% correct on a test' usually would.

- 150% of a goal makes sense because ____.
- 150% on a test does not make sense because ____.

EXPLORE

When you wrote percents like 150% and 0.5% as decimals, what did you notice about their size compared to 1?

Level 1 support

Start here: Percents over 100% become decimals bigger than 1, and percents under 1% become decimals close to 0.

 Need a hint?

Sentence starters (optional)

150% as a decimal is ____ because ____.

150% como decimal es ____ porque ____.

Percents over 100% have decimals ____ than 1, and percents under 1% have decimals ____.

Los porcentajes mayores que 100% tienen decimales ____ que 1, y los menores que 1% tienen decimales ____.

Word bank: **percent** **equivalent** **greater than 100%** **less than 1%** **mixed number**

Listen for: Students state $150\% = 1.5$ (greater than 1) and $0.5\% = 0.005$ (much less than 0.01), and recognize these forms are equivalent.

Level 2

Why does 0.5% turn into 0.005 and not 0.05? Explain the role of dividing by 100, and write 150% as a mixed number too.

- 0.5% becomes 0.005 because ____.
- 150% as a mixed number is ____ because ____.

CONNECT

A jackpot pays 250% of your tokens and you inserted 40 tokens. One friend says 100 tokens, another says 60. Who is right, and why does '250%' mean more than your tokens?

Level 1 support

Start here: 250% of 40 is 2.5 times 40, which is 100 tokens.

 **Need a hint?**

Sentence starters (optional)

The friend who said ____ tokens is correct because 250% = ____ times the whole.

El amigo que dijo ____ fichas tiene razón porque 250% = ____ veces el entero.

250% of 40 = ____ x 40 = ____.

250% de 40 = ____ x 40 = ____.

Word bank: percent greater than 100% equivalent mixed number

Listen for: Students convert 250% to 2.5, multiply $2.5 \times 40 = 100$, and explain that over 100% means more than the original 40 tokens.

Level 2

The friend who said 60 found 150% of 40. Explain the mistake they likely made, and describe a jackpot percent that WOULD give exactly 40 tokens.

- The 60-token friend probably ____.
- You would get exactly 40 tokens when the jackpot is ____% because ____.

Write About the Math

The Writing Revolution

I can explain my reasoning using the words percent, greater than 100%, less than 1%, and equivalent.

1. Kernel Sentence subject + verb

Model: Percent is a way to compare a number to 100, shown with the % sign.

Porcentaje es una manera de comparar un número con 100, con el signo %.

Write a kernel sentence about percent. Use a subject and a verb.

Escribe una oración base sobre porcentaje. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Percent matters in math

Porcentaje importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Percent matters in math because ____.

Porcentaje importa en matemáticas porque ____.

but
pero

Percent matters in math, but ____.

Porcentaje importa en matemáticas, pero ____.

so
entonces

Percent matters in math, so ____.

Porcentaje importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about percent.
Di un hecho verdadero sobre percent.

Percent ____.

Question *Pregunta*

Ask a question about percent.
Haz una pregunta sobre percent.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about percent.
Muestra entusiasmo sobre percent.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with percent.
Dile a un compañero qué hacer con percent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

____ % means ____ because it is more / less than 100%.

____ % significa ____ porque es más / menos que 100%.

A percent over 100% means ____, and a percent under 1% means ____.

Un porcentaje mayor que 100% significa ____, y uno menor que 1% significa ____.

150% as a decimal is ____ because ____.

150% como decimal es ____ porque ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. A price rises to 120% of its original value. The original was \$40. What is the new price?

- A. \$48
- B. \$44
- C. \$60
- D. \$52

Show your work:

2. Which is the smallest value?

- A. 0.2%
- B. 2%
- C. 20%
- D. 0.2

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A game character has 300% of its original health after picking up a power-up. If the original health was 80 points, how many health points does the character have now? A student says 240 points. Another says 380 points. Who is correct and why?

Sentence starter: The student who said ____ is correct because 300% means ____ times the original, so ____ $\times 80 =$ ____.

Show your work:

Reflect — Exit Ticket

Which statement is true about 0.25%?

- A. It equals 0.0025 as a decimal
- B. It equals 0.25 as a decimal
- C. It equals $\frac{1}{4}$ as a fraction
- D. It is greater than 1%

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $2.0 - 200\% \div 100 = 2.0$. *Percents greater than 100% become decimals greater than 1.*
2. **You Try (notes):** A. $0.005 - 0.5\% \div 100 = 0.005$. *Move the decimal two places to the left.*
3. **Try It 1:** A. $\$48 - 120\% \text{ of } 40 = 1.2 \times 40 = 48$.
4. **Try It 2:** A. $0.2\% - 0.2\% = 0.002$, *smaller than 2% (0.02), 20% (0.2), and 0.2.*
5. **Exit Ticket:** A. It equals 0.0025 as a decimal — $0.25\% = 0.25 \div 100 = 0.0025$. *It is much less than 1%, not equal to 1/4.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Percent is a way to compare a number to 100, shown with the % sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).